



PRIFYSGOL
BANGOR
UNIVERSITY

Assignment II – Design and Evaluation of an Improved Computer Interface

Scenario Chosen: Smartwatch Application to Complement MyBangor

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Module: ICE-4141-0 User Experience (UX) & HCI

Table of Contents

Introduction	4
1. Task 1 – Market Research	4
1.1 MyBangor Student Portal.....	4
1.2 Todoist (Apple Watch)	5
2.3 Class Timetable (Apple Watch)	6
2.4 Market Research Summary	7
2. Task 2 – Content Outline	7
3. Task 3 – Storyboarding	7
4. Task 4 – Wireframing.....	8
5. Task 5 – Mock-up.....	11
6. Conclusion	12
7. References	13



Gen-AI in assessment declaration

Student 500 number: 500750651

Module Code: ICE-4141

Assessment: Category 2

Assessment's Gen-AI Category (see the Assessment Brief): 2

The following statement reflects my use of Gen-AI within this assessment (select all that apply):

Description of use	Select (X)
Gen-AI has not been used at all in producing this assessment.	X
Gen-AI was used in an assistive role to research topics and provide explanations.	X
Gen-AI was used in an assistive role to suggest sources of information, e.g. publications to read.	X
Gen-AI was used in an assistive role to summarize the content of publications or other sources.	
Gen-AI was used in an assistive role to produce an outline or draft structure of my assessment.	
Note: for Category 2 assessments, the fundamental content of the submission (e.g., the arguments, interpretations, conclusions etc) must be your own.	
Gen-AI was used in an assistive role to improve spelling, grammar, sentence structure and/or clarity of expression of my assessment.	
Note: for Category 2 assessments, the basic written expression must be your own. So <u>Therefore</u> whilst Gen-AI may correct mistakes it must not produce the actual content.	
Gen-AI was used in an assistive role to translate text during the production of the work.	
Gen-AI was used in an assistive role to analyse data and/or produce visualizations of data.	
Gen-AI was used in an assistive role to help produce creative outputs e.g. art, imagery, movie, music, creative writing.	
Gen-AI was used in an assistive role to refine computer code and/or other output.	
Gen-AI was used in another assistive role (please describe):	
Gen-AI was fully used to complete a Category 3 assessment and use complied with the assessment instructions provided.	

By using this cover sheet and entering my student ID above, I declare that I have provided honest and accurate details regarding my use of Gen-AI within this assessment.

Introduction

The purpose of this assignment is to demonstrate understanding of Human–Computer Interaction (HCI) and User Experience (UX) principles through the structured design and prototyping of an interface. The work follows guidance provided and applies established HCI techniques across multiple stages of the design process.

The chosen scenario is a smartwatch application intended to complement the MyBangor student portal. The application is not designed to replace the existing system, but to act as a companion interface that enhances student engagement by providing quick, glance-based access to academic information. Smartwatch interactions are typically brief and context-driven, making the platform suitable for surfacing information such as upcoming classes, assignment deadlines, and important notifications.

This report documents the complete design process, progressing from market research and content definition to storyboarding, wireframing, and high-fidelity mock-up creation. Each design decision is justified using recognised HCI principles to ensure usability, clarity, and suitability for wearable devices.

1. Task 1 – Market Research

This task evaluates three relevant applications to identify good and bad UX practices in a smartwatch design., the MyBangor student portal and two Apple Watch applications that demonstrate effective wearable interaction patterns. The focus is on information prioritisation, clarity, and ease of access.

1.1 MyBangor Student Portal

Developer: Bangor University

Platform: ChromeOS 143.0.7499.41

URL: <https://my.bangor.ac.uk>

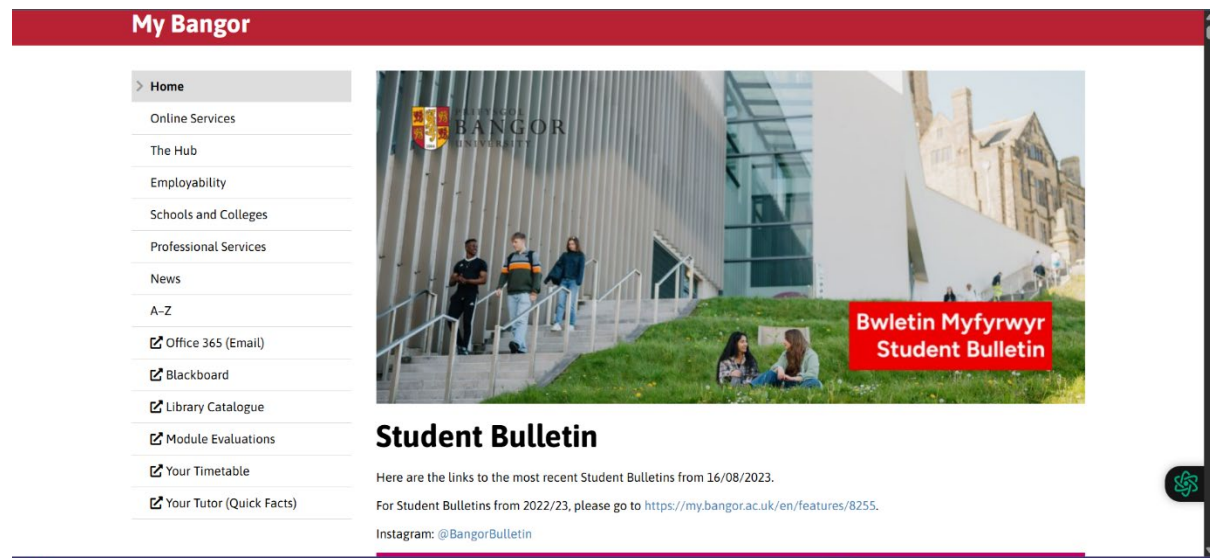
MyBangor is the primary digital platform through which students access academic information such as timetables, assessments, announcements, and support services. A key strength of the portal is the centralisation of services, providing a single central system that students can rely on. Content is personalised based on enrolment, improving relevance.

However, the portal presents a large volume of information simultaneously. Time-critical content such as assignment deadlines and upcoming classes often appears alongside general notices, reducing its visual prominence (which is accessed through MyBangor). This can make it difficult for students to quickly identify what requires immediate attention.

For a smartwatch context, MyBangor highlights the need for selective information presentation. A companion smartwatch application can address these limitations by filtering and prioritising only the most urgent academic information and presenting it in a simplified, glance-based format.

In addition to individual application analysis, a comparison across the reviewed systems highlights clear design trade-offs. MyBangor prioritises completeness and institutional coverage but sacrifices immediacy, while Todoist and Class Timetable emphasise rapid access at the cost of contextual depth. This contrast reinforces the suitability of a smartwatch companion that bridges these

approaches by combining academic context with glance-based prioritisation. The smartwatch interface can therefore act as a filtering layer, surfacing only the most relevant elements of the MyBangor ecosystem while avoiding the information overload present in the full portal.



Screenshot of MyBangor homepage illustrating information density.

1.2 Todoist (Apple Watch)

Developer: Doist Inc.

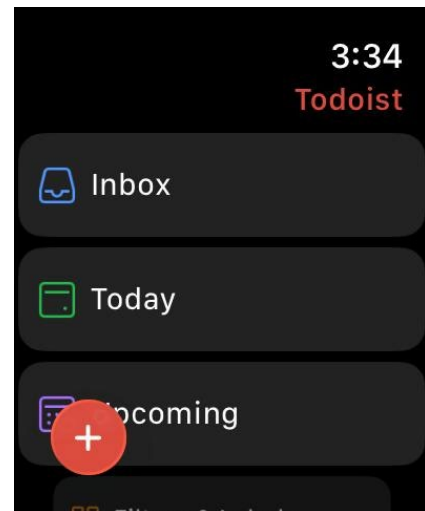
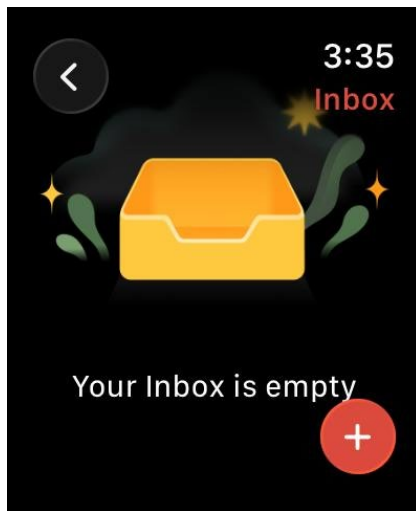
Platform: watchOS 26.2

URL: <https://apps.apple.com/gb/app/todoist-to-do-list-calendar/id585829637>

Todoist is a widely used task-management application with Apple Watch support. Its smartwatch interface focuses on displaying upcoming tasks and deadlines using concise text and minimal interaction, allowing users to assess priorities quickly.

A limitation of Todoist in an academic context is the lack of domain-specific information. Tasks are generic and not explicitly linked to modules or institutional systems. Some interaction elements are also small, which may increase the risk of accidental input on a smartwatch. This also varies and depends on the screen size as it is mostly narrow and limited. So, the information provided and interaction should be concise and minimal.

Todoist demonstrates effective deadline visibility and prioritisation on wearable devices. These patterns can be adapted to an academic setting by combining deadline awareness with contextual information drawn from MyBangor.



Screenshots of Todoist Apple Watch interface.

1.3 Class Timetable (Apple Watch)

Developer: Class Timetable LLC

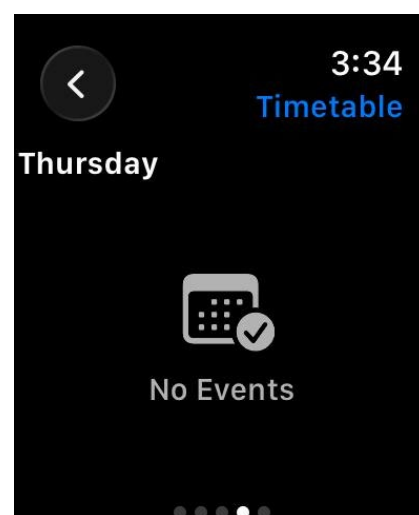
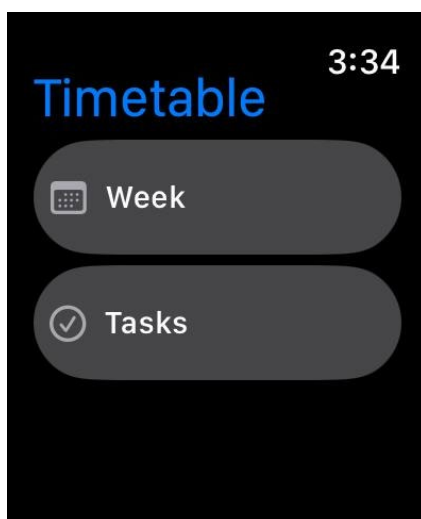
Platform: watchOS 26.2

URL: <https://apps.apple.com/gb/app/class-timetable/id425121147>

Class Timetable is a student-focused scheduling application that provides simplified access to class schedules and reminders, including support for Apple Watch. Its strength lies in presenting schedule information clearly for quick reference.

However, displaying excessive schedule detail on a small screen can lead to clutter if not carefully prioritised. Task and deadline features are relatively basic and lack deeper academic context.

This application demonstrates how student schedules can be distilled effectively for smartwatch use. It supports the design decision to prioritise “today’s schedule” or “next class” information rather than full weekly timetables.



Screenshots of Class Timetable Apple Watch view.

1.4 Market Research Summary

The reviewed systems show that effective smartwatch UX relies on glance-based prioritisation, minimal interaction, and clear urgency cues. These insights directly confirm to the MyBangor smartwatch design, which focuses on timetable awareness, assignment deadlines, and notifications to support student engagement.

2. Task 2 – Content Outline

The smartwatch application content is deliberately constrained to support quick interaction and reduced cognitive load. Content is grouped into logical categories aligned with smartwatch usage patterns.

Overview (Main View): Displays the next scheduled class, the nearest assignment deadline, and any high-priority alerts. This supports recognition over recall and immediate awareness of time-critical information.

Timetable (Daily View): Shows only the current day's classes with start times and a countdown to the next session. This avoids information overload while maintaining temporal awareness.

Assignments and Deadlines: Lists upcoming assessments with due dates, submission status, and visual urgency indicators. Visualising urgency supports prioritisation and student engagement.

Notifications and Alerts: Centralised view of deadline reminders, timetable changes, and important announcements. Push-based alerts reduce the need for repeated manual checking.

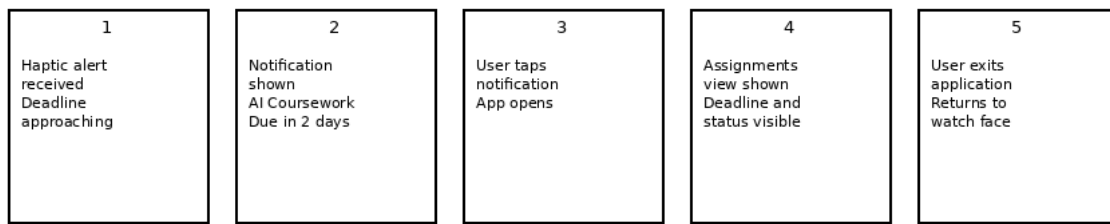
Personalisation and Settings: Allows control of notification preferences and display mode. This supports individual needs without introducing unnecessary complexity.

3. Task 3 – Storyboarding

Chosen interaction: Viewing an upcoming assignment deadline from a smartwatch notification.

Storyboard description:

- The student receives haptic feedback and a ping alert indicating an upcoming assignment deadline.
- The notification displays the assignment name and remaining time.
- The student taps the notification, opening the Assignments view.
- The assignment deadline and submission status are displayed.
- The student exits the application and returns to the watch face.



Storyboard illustrating the interaction sequence.

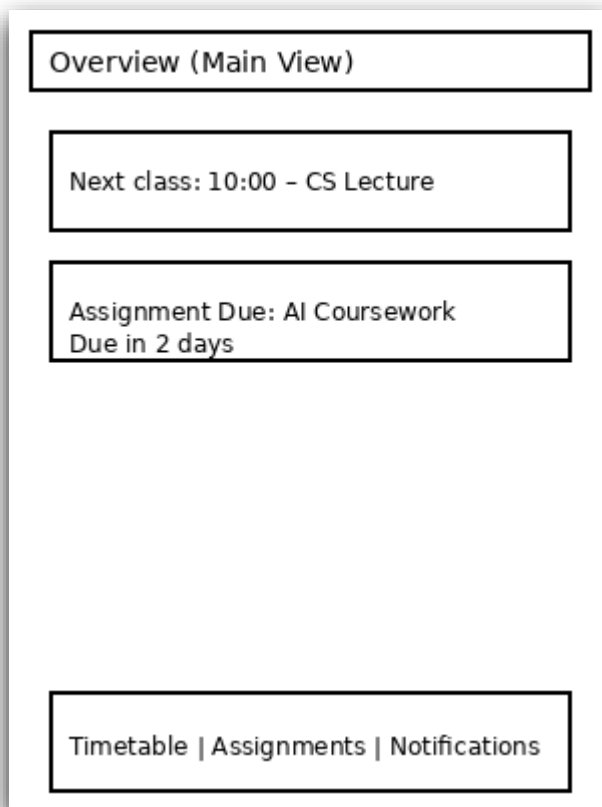
Conformance to Shneiderman's Golden Rules:

This interaction provides immediate feedback through haptic and visual cues, ensuring visibility of system status. Consistency is maintained across screens through uniform layout and indicators. Recognition is prioritised over recall by presenting deadlines. Cognitive load is reduced by limiting information per screen, while user control is preserved through simple exit gestures. Error prevention is supported by avoiding complex input, and clear closure is achieved once the information has been reviewed.

The storyboard also aligns with the principle of consistency by maintaining predictable navigation and visual structure across each step of the interaction. By limiting the number of actions required to complete the task, the interaction supports efficiency of use and accommodates users under time pressure. The absence of text entry or complex gestures further reduces the likelihood of user error, which is particularly important in smartwatch contexts where interaction precision is limited. Overall, the storyboard demonstrates how HCI rules can be operationalised in a realistic academic use case rather than remaining abstract guidelines.

4. Task 4 – Wireframing

Three wireframes were created, each representing a separate page of the interface. One wireframe function as the main view. The wireframes focus on layout, hierarchy, and navigation rather than visual styling.



Wireframe 1 – Overview screen (main view).

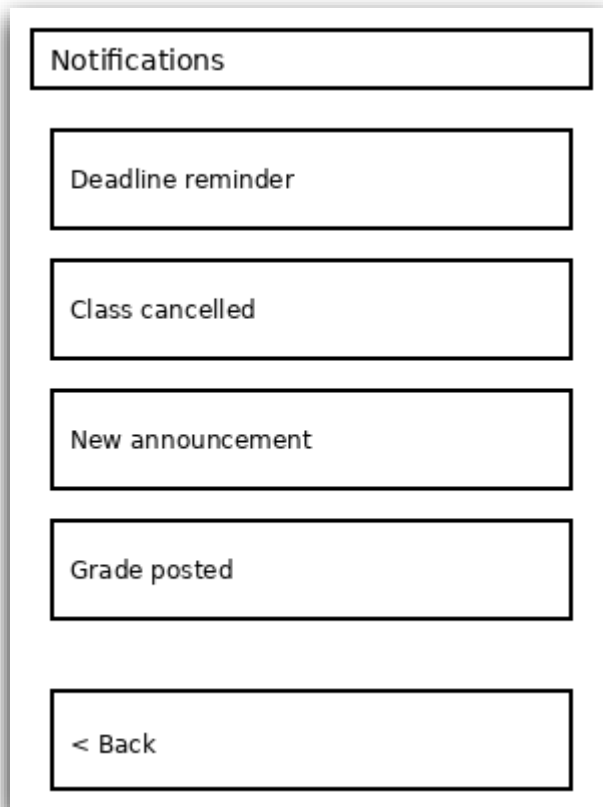
The Overview wireframe establishes a clear hub layout. Visual hierarchy emphasises urgent information, while proximity groups related academic elements. The minimalist structure reduces cognitive load and supports glance-based interaction.

The wireframe illustrates a mobile application screen titled 'Assignments'. It features a vertical list of four assignment items, each in its own rectangular box. The items are: 'AI Coursework - Due in 2 days', 'Group Project - Due in 7 days', 'Lab Report - Submitted', and 'Research Paper - Due in 5 days'. At the bottom of the screen is a separate box containing a back button labeled '< Back'. The entire screen is enclosed in a light gray border with a subtle drop shadow.

Assignments
AI Coursework - Due in 2 days
Group Project - Due in 7 days
Lab Report - Submitted
Research Paper - Due in 5 days
< Back

Wireframe 2 – Assignments screen.

The Assignments wireframe uses a vertical list layout that supports continuity and efficient scanning. Similar list items reinforce consistency and visibility of system status without enabling error-prone actions.



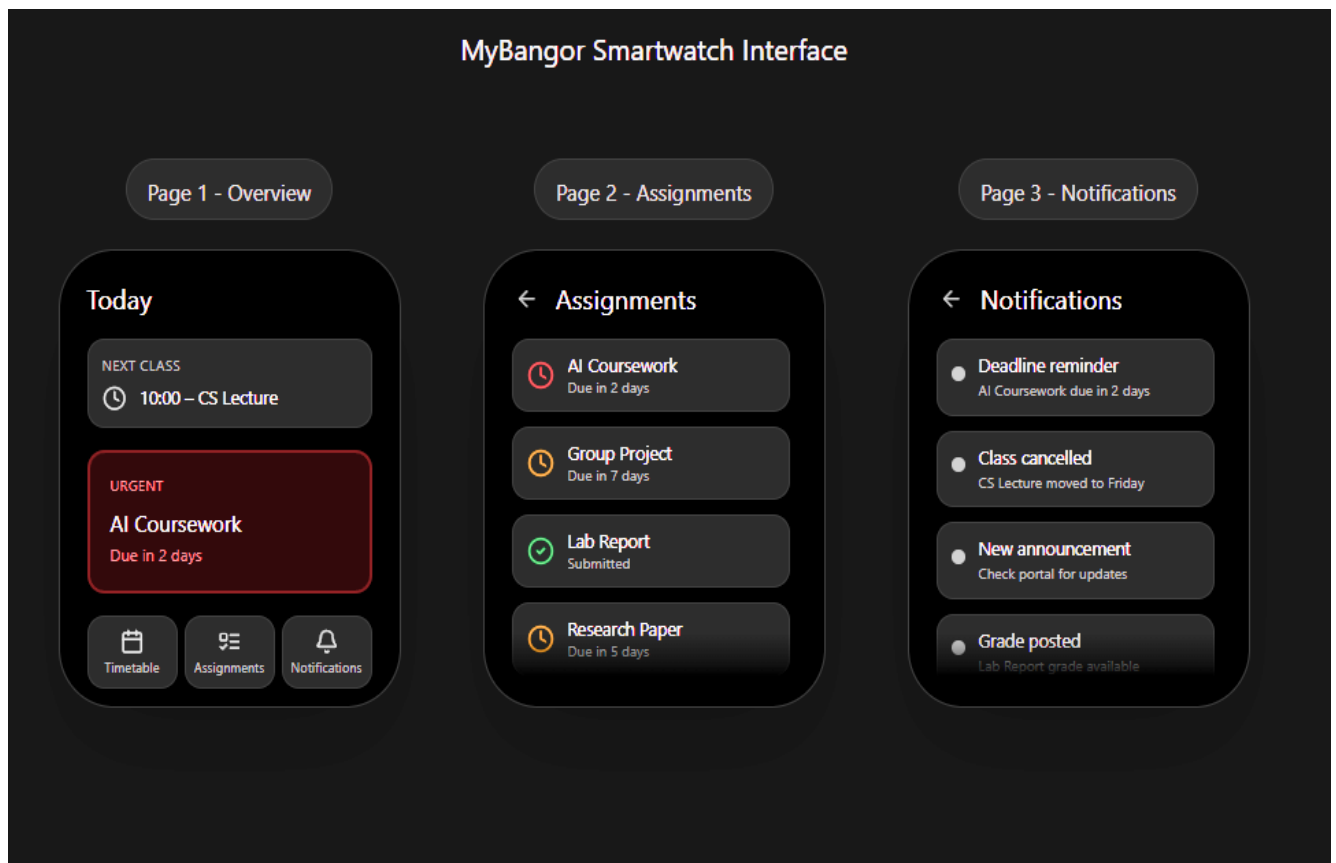
Low-fidelity wireframe – Notifications screen.

The Notifications wireframe applies closure and proximity to group alerts clearly. Navigation remains consistent across screens, supporting user control and predictability.

Across all three wireframes, consistency is maintained in layout structure and navigation placement, supporting learnability and reducing the need for user training. The repeated use of similar list structures and headers applies the Gestalt principle of similarity, enabling users to transfer understanding from one screen to another. Additionally, the use of whitespace and clear separation between content blocks supports the principle of proximity, helping users perceive related information as coherent groups. These structural decisions are particularly important for smartwatch interfaces, where limited screen real estate demands careful organisation to avoid cognitive overload.

5. Task 5 – Mock-up

A high-fidelity mock-up consisting of three smartwatch views was created using Figma, directly based on the wireframes developed in Task 4. The mock-up refines visual presentation while preserving layout and navigation structure.



High-fidelity smartwatch mock-up showing Overview, Assignments, and Notifications views

Colour scheme and visual strategy:

A dark background was selected to align with smartwatch conventions and improve legibility during brief interactions. Accent colours are used strictly for functional status, with red indicating urgent deadlines, amber indicating upcoming deadlines, and green indicating completed tasks. Primary text uses light colours for contrast, and typography is clean and legible. The overall strategy follows aesthetic and minimalist design principles, ensuring that each visual element serves a functional purpose.

Accessibility considerations were also considered in the visual design. High contrast between text and background supports readability in varying lighting conditions, while colour is not used as the sole means of conveying information. Status indicators are reinforced through text labels and spatial positioning, ensuring that users with colour-vision deficiencies can still interpret system state accurately.

6. Conclusion

This assignment demonstrates a structured application of HCI and UX methods to the design of a smartwatch application that complements the MyBangor portal. Through market research, content definition, storyboarding, wireframing, and high-fidelity prototyping, the design addresses limitations of the existing portal while respecting wearable constraints. The resulting interface enhances student engagement by prioritising clarity, urgency, and usability.

7. References

Bangor University (2025). MyBangor Student Portal. Available at: <https://my.bangor.ac.uk>

Doist Inc. (2025). Todoist (Apple Watch application). Available at: <https://apps.apple.com/gb/app/todoist-to-do-list-calendar/id585829637>

Class Timetable LLC (2025). Class Timetable (Apple Watch application). Available at: <https://apps.apple.com/gb/app/class-timetable/id425121147>

Figma Inc. (2025). Figma: Interface design and prototyping tool. Available at: <https://www.figma.com>

Shneiderman, B. (1992). Designing the User Interface.

draw.io (2025). draw.io Diagramming Tool. Available at: <https://www.draw.io> – used for storyboarding and wireframing.